INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 6: Conclusion and Future Work

by

TAEHONG KWON

Student number: 28576810

submitted in accordance with the requirements for the degree of

DOCTOR OF PHILOSOPHY

in the subject

COMPUTER SCIENCE

at the

UNIVERSITY OF SOUTH AFRICA

SUPERVISOR: Prof. Ernest Mnkandla

CO-SUPERVISOR: Prof. Donatien Koulla Moulla

CO-SUPERVISOR: Dr. David Sena Attipoe

October 2026

6 Conclusion and Future Work

Chapter 5 interpreted the findings and judged whether the study met its objectives. The tables of Chapter 4 form the record.

6.1 Answers to the research questions

The thesis built a coupled PageRank (C-PR), one walk over ERC-20 allowance and transfer edges mixed at each address with a fixed weight λ, and evaluated it beside its single-layer references, EndorseRank (PageRank with the edge weights of Adaptive Weighted PageRank, AWP, on the latest allowances) and AWP itself (T. Do et al., 2023), its ablation S-PR and the raw degrees, on Arbitrum One contracts from October 2023 to June 2026, with holdout windows W0 (April to June 2026) and W1 (July to September 2026).

RQ1 asked whether EndorseRank and AWP, the walks over allowance edges alone and over transfer edges alone, rank the contracts of one sample differently, and how closely each score agrees with allowance and transfer statistics in the same window. They rank the contracts differently, and H1 is supported, largely by construction (Section 4.4). Each score follows the edges it walks on (Table 4.3), which supports H2 only narrowly, because the proxies come from the same events as the graphs. The divergence is the premise of the coupled walk.

RQ2 asked what each score costs to compute, and how much walking both edge sets adds. The cost follows the edge count, so EndorseRank, on the sparsest graph, is the cheapest score, and the coupled walks cost more than AWP (Table 4.1); H3 is supported, largely by construction, and on subsamples only at the two largest stages (Table 4.2).

RQ3 asked whether scores frozen at t_1 predict the new approval pairs and new transfer senders that spender contracts gain afterwards, and whether any PageRank, the coupled one included, predicts

them better than the raw degree of the same edge type. The first part holds in both windows, and the edge type decides which label a score anticipates (Tables 4.9 and 4.15). The second differs by walk, in contrasts computed after the labels were known: AWP ranks later new senders above the transfer in-degree in both windows, EndorseRank ranks later approval pairs below the in-approve degree in both (Tables 4.10 and 4.13), and C-PR ranks neither label above the degree of the same edge type (Table 4.18). EndorseRank with AWP's activity restarts ranked later approval pairs above the degree in both windows, a post hoc hypothesis, not a finding.

RQ4 asked whether C-PR, the method of the thesis, ranks contracts better than either of its layers walked alone. The answer is no, on the revised scores as in the registered analysis (Table 8.12). Under rule A, C-PR meets the allowance condition of the primary part but fails its transfer condition and the secondary part (Section 4.7.1; Table 4.10). The seeded ablation matches the transfer walk on new senders: allowance edges contribute when the walk moves along them, and not as a seed.

RQ5 asked whether adding the allowance layer improves a walk over the transfers alone. The answer is again no. Under rule B, the exploratory gain over the transfer walk on new approval pairs in W0 recurred in W1, but the loss on new senders exceeded the margin, and on the traders' liquidation-free close rate a loss beyond it could not be ruled out; against AWP the loss on new senders was clear (Table 4.16). Beside the rules, in contrasts computed after the labels were known, C-PR ranks both spender labels above EndorseRank in both windows, its lead over AWP on later approvals comes from the contracts that AWP scores zero, and it ranks later approvals below EndorseRank with activity restarts (Table 4.18).

In a comparison registered with rule B that adds no research question (Section 5.1.6), the approval count at the freeze led the transfer count on later liquidation avoidance, the outcome built from neither edge type that bears on counterparty risk, in W0 but not resolvably in W1, the primary window ($P = +0.084$, 97.5% interval $[-0.026, 0.188]$; Section 4.8). The like-for-like PageRank pair is not resolved, the signal concerns forced-liquidation avoidance, not trading success, and the samples are small.

Claims C1–C6 (Section 4.9) state these findings formally.

The main research question asked what one walk over both edge types adds to ranking these contracts. Against the walk over allowances alone, the coupled walk adds the transfer construct and ranks every later spender label higher; against the walk over transfers alone, it adds delegated authority, which ranks later approval pairs higher only through the contracts that walk cannot rank, at a loss on later inflow. Under the rules fixed in advance it beats neither layer, and neither it nor the allowance walk with uniform restarts beat the raw degree of the same edge type out of window.

6.2 Original contribution to knowledge

In the terms of the literature reviewed in Chapter 2, the five contributions of Section 1.9 establish the following.

Before this work, graph reputation on public ledgers ranked wallets by their transfers (H.-D. Do & Do, 2023; T. Do et al., 2023), by how risk spreads through transaction networks (D. Lin et al., 2025), or by explicit ratings (Almasoud et al., 2020; Bellini et al., 2020), and the reputation tooling deployed in DeFi scored the holder of an address from credentials (Human Passport, n.d.). None of it used the ERC-20 allowance, the on-chain record of a deliberate delegation of spending power (Vogelsteller & Buterin, 2015), as the edge of a reputation ranking, coupled it with payments in one walk, or ranked contract accounts by the approvals they receive; approvals had been used to cluster the addresses of one entity (Victor, 2020) and studied as a security risk (Wang et al., 2022). None evaluated on-chain reputation scores on one cohort against the checks of each construct and against the raw degrees they smooth, with confidence intervals, difference tests and temporal holdouts under decision rules fixed in advance. This thesis supplies both. PageRank and AWP's weighting are taken from the literature (T. Do et al., 2023; Page et al., 1999); the coupled walk changes the transition rule, and EndorseRank, its allowance reference, changes only the edge.

The new knowledge has five parts. First, what an on-chain reputation score of a contract measures depends on the edges it uses, and this can be checked on one cohort with quantified uncertainty. Second, a walk that couples the allowance edge with transfers, a step that the multilayer literature

suggested, beats neither of its layers under rules fixed in advance; beside the rules, it ranks every later spender label above the walk over allowances alone and adds to the walk over transfers an order among the contracts that walk leaves tied, at a loss on later inflow. Third, the allowance edge alone yields a score that is cheaper to update, differs from the transfer-based score and ranks later endorsements of spenders in two windows; on few traders, the comparison registered with rule B found no count-level lead of allowance over transfer on later avoidance of forced liquidation in the primary window. Fourth, propagation beat the raw degree it smooths only on transfer edges, in AWP; neither EndorseRank nor the coupled walk beat the count of its edges. Fifth, at depth one a walk with uniform restarts is, up to a constant, a linear function of the transition mass into a spender, which explains much of the allowance-side result (Section 5.3.2). The construct-validity framing (Cronbach & Meehl, 1955; Messick, 1995) that makes these claims testable, and the open pipeline whose published summaries let others check every table, are part of the contribution, since whether decentralized systems miscalculate credit scores is a matter of public interest (Packin & Lev-Aretz, 2024).

The scope is deliberately narrow. The thesis does not show that on-chain reputation can replace collateral (Gudgeon, Perez, et al., 2020; Schär, 2021), that the results apply beyond contracts approved on the five tokens on Arbitrum One in the window studied, or that any score measures creditworthiness, default risk or trading skill, and it does not claim that C-PR or EndorseRank resists Sybils: under uniform teleportation a closed farm of attacker addresses can inflate either cheaply (Section 5.6).

6.3 Future work

Each suggestion below corresponds to a limitation identified in Chapter 5. Any new score or weighting it leads to should be fixed before it is scored, replicated on registered new data and checked out of window against degree baselines.

- A registered test of the coupled walk. A registration committed before the logs of a later

window, such as October to December 2026, are extracted would fix the revised C-PR with its US-dollar amounts, five tokens, value slopes and allowance cap, together with a coupled walk whose layers carry the edges and weights of AWP and EndorseRank and whose restarts mix sending and approving activity (C-AWP, whose ends are AWP and EndorseRank with activity restarts), with the cohort rule, rules A and B and the comparison on traders, which needs more contract traders than W0 and W1 held. C-PR differs from AWP in its coupling, its weights and its restarts at once; C-AWP would set the coupling alone against the published walks. The value-slope check did not cover the holdout labels (Table 4.6), so the slopes and the cap should not be tuned on the new window. Node-dependent or learned weights, per-layer damping and mixtures of the two stationary vectors are further couplings that this thesis does not evaluate; any of them should keep the auditability that sets graph propagation apart from opaque learned scores (Kotb, 2023; Packin & Lev-Aretz, 2024).

- Propagation on a deeper sample. On the ego networks studied, EndorseRank reduces to a reweighted approval count for every depth-one spender (Section 5.3.2); a sample extracted two or more hops out, with the variant with AWP's activity restarts tested against the in-approve degree on new data separately for depth-one and deeper spenders, would show whether propagation, and not the restarts alone, adds to the count.
- Validate the results in other contexts. Running the same analysis on other rollups, on Layer-1 networks and over later windows would show whether the results extend beyond Arbitrum One and the months studied; because DeFi is heterogeneous (Schär, 2021; Werner et al., 2022), this is the main threat to external validity.
- Adversarial evaluation. The Sybil cost model is analytical, and no attack was carried out on chain (Section 5.6). Restricting teleportation and dangling redistribution to a vetted seed set, as TrustRank does (Gyöngyi et al., 2004), would remove the uniform inflow that closed farms exploit, and red-teaming with wash approvals, coordinated spender farms and strategic revocation would measure the realized cost of attacks and test whether the Sybil-adjusted proxies remain informative.

- Composition with attestation tooling. Filtering the approving owners through a Passport-style gate (Human Passport, n.d.) before C-PR or EndorseRank orders the contracts is a plausible deployment; it should be evaluated on labeled Sybil data, with the gate's own weaknesses (Bordeianu & Popescu, 2026; Siddarth et al., 2020) in view.
- Deployment studies. Integrating the scores with risk oracles and governance tooling would test whether their runtime profiles can sustain production refresh cadences. Controlled experiments on fee tiers or monitoring cadence would let evidence replace the adoption hypotheses of Section 5.4.4.

6.4 Closing reflection

In adversarial, pseudonymous markets on-chain reputation does not remove the need for collateral (Gudgeon, Perez, et al., 2020), but it can make behavioral structure readable, and which structure it reads depends on the edges used. A transfer records economic flow and an allowance records delegated authority. PageRank will rank either graph, and one walk can follow both. On these data the coupled walk ranked every later spender label above the walk over allowances alone, but it beat neither layer under the rules fixed in advance and gained on the walk over transfers only among the contracts that walk cannot rank. For a protocol or researcher who needs a fast, auditable signal of which contracts owners trust, the allowance edge is the one to use on data like those studied here. The approval count is its cheapest form and ranked later approvals better than EndorseRank with uniform restarts and, in W1, than C-PR; EndorseRank is the propagated form, exposed to cheap inflation by closed farms (Section 5.6). For describing transfer hubs and anticipating their inflow, the transfer edge is the one to use, and there the walk improved on the count. For screening counterparties against forced liquidation, the approval count led the transfer count among contract traders in W0 but not resolvably in the primary window, and more traders are needed to settle it.